Utility Municipal Bonds and Wildfire in the Western United States

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Introduction

In a world increasingly shaped by climate-related hazards, understanding the financial consequences of these risks is critical – particularly for public utilities that underpin essential infrastructure. Among these hazards, wildfire risk poses an acute and growing challenge for utilities in the Western United States. Its impact extends beyond physical damage, influencing the financial viability and cost of capital for local governments that rely on municipal bonds to fund infrastructure and energy projects.

This study examines how escalating wildfire risk affects access to capital for public utility providers by analyzing its influence on municipal bond pricing. A foundational contribution to the literature on climate risk and municipal finance is the work of Keenan and Kahn [3], who find that climate change exposure – particularly sea level rise – is increasingly priced into municipal bond yields. Their analysis shows that communities facing higher projected physical risks tend to pay higher borrowing costs, even after controlling for economic and demographic characteristics. This finding underscores the growing salience of environmental risk in credit markets. Our study builds on this insight by shifting the focus to wildfire risk and examining its impact on public utility district bonds. Unlike sea level rise, which unfolds slowly over time, wildfires present sudden and acute hazards that may have more immediate implications for infrastructure and municipal creditworthiness. Such models can support utilities in managing their long-term financing strategies while helping investors more accurately evaluate the creditworthiness of issuers operating in fire-prone regions. In doing so, this approach offers a pathway for integrating climate risk into fixed income analysis, with implications for both resilience planning and sustainable investing.

Literature Review

Recent research has increasingly explored how climate-related risks are capitalized into municipal bond markets. One of the most relevant studies in this space is by Jeon, Barrage, and Walsh, Pricing Climate Risks: Evidence from Wildfires and Municipal Bonds [2], who investigate how projected wildfire risks influence borrowing costs for U.S. school districts. Their analysis leverages high-resolution drought projections, land use patterns, and municipal bond transaction data to estimate the relationship between expected wildfire exposure and bond spreads. Specifically, they focus on general obligation bonds issued by school districts and model yield spreads as a function of interpolated changes in wildfire potential, using the Keetch–Byram Drought Index (KBDI) as their primary wildfire risk metric.

KBDI is a drought index that estimates soil moisture deficiency based on weather-driven inputs such as daily maximum temperature, recent precipitation, and long-term rainfall accu-

mulation. The index, ranging from 0 to 800, is commonly used by fire management agencies to assess wildfire potential under meteorological conditions [4, 1]. In their methodology, Jeon et al. [2] combine spatially weighted KBDI values with Wildland-Urban Interface (WUI) data to account for the distribution of exposed housing units. These values are then matched to school districts based on bond maturity and projected mid-century climate conditions. Their findings reveal that a one standard deviation increase in projected KBDI corresponds to a 23 basis point rise in bond spreads after 2014 – an economically significant effect, especially in districts with higher minority populations or greater reliance on local revenue.

While our research is conceptually aligned with Jeon et al. [2] in its focus on the fiscal consequences of wildfire risk in fixed income markets, we introduce several methodological distinctions. First, rather than focusing on school districts and general obligation bonds, we analyze revenue bonds issued by public utility districts in the Western United States, which often finance critical electric infrastructure. Second, instead of relying solely on interpolated KBDI forecasts, we incorporate two complementary sources of fire risk: (1) observed historical wildfire perimeters from the Monitoring Trends in Burn Severity (MTBS) dataset, and (2) seasonal Fire Weather Index (FWI) data derived from GEOS-5. FWI, developed by the Canadian Forest Service, incorporates additional meteorological variables – such as wind speed and relative humidity – not captured by KBDI, and is increasingly used in global climate fire models [1].

By integrating both historical fire footprints and dynamic forward-looking weather indices, our approach offers a more granular perspective on how fire exposure – both realized and anticipated – may affect municipal credit risk. This dual focus also allows us to test whether capital markets respond differently to past fire occurrences versus projected climatic hazards, and to assess the robustness of pricing responses across bond types and issuer categories.

Data Source

We primarily used FactSet as the source for municipal bond data. Our analysis focuses on three states on the West Coast – Washington, Oregon, and California – due to their increased exposure to wildfire risk. We began with the Starter Screen – Municipal in FactSet and customized the dataset by adding an “Issuer Name” column to facilitate future spatial matching with the GeoJSON file of the Electric Retail Service Territories, which we extracted from the U.S. Energy Atlas.

The focus on electric utility service areas reflects the critical vulnerability of power infrastructure to wildfires. Wildfire events can cause significant damage to transmission and distribution networks, interrupt power supply, and lead to substantial financial and legal liabilities for utilities. Wildfire events pose increasing liability for utility infrastructure own-

ers and operators [5], and therefore, understanding how exposure to fire risk translates into financing costs for electric utilities is essential for both investors and policymakers.

By matching bond issuers to their corresponding electric utility service areas, we enhance the spatial precision of our analysis. This geographic alignment allows us to assess how specific service territories – rather than broad municipal boundaries – have been historically impacted by wildfires, and in turn, how these exposures may influence credit spreads. Such matching enables more reliable forecasting of future borrowing costs under evolving climate risk scenarios.

To ensure data quality, we restricted our dataset to non-default municipal bonds, avoiding potential distortions in modeling outcomes. We then filtered the data to include only bonds whose issuer names contain *Pub Util Dist*, aligning with our research goal of evaluating wildfire impacts on bonds issued by public utility districts.

The spatial matching between FactSet bond issuers and electric service territories was performed using a large language model (LLM) to identify name similarities, followed by manual verification. Although this process enhanced matching coverage, we recognize that it may introduce uncertainties if some alignments were only approximate.

We also supplement our analysis with daily municipal yield curve data from Tradeweb, which provides benchmark yields for municipal bonds across maturities ranging from 1 to 30 years. This dataset offers a high-frequency view of borrowing costs in the tax-exempt bond market and serves as a market-based reference for evaluating the pricing of municipal credit risk. We use this data to contextualize yield trends over time and explore whether wildfire risk exposure may influence the term structure of municipal borrowing rates.

In addition to financial data, we integrated two sources of wildfire risk information:

- Historical observed FWI values from https://data.giss.nasa.gov/impacts/gfwed/ , specifically the GEOS-5. This way we were able to see the actual FWI values the day the fire was ignited,
- Monitoring Trends in Burn Severity (MTBS), which provides spatial data on historical wildfire perimeters and severity.

Together, these datasets form the foundation of our analysis and support a multifaceted examination of how wildfire exposure affects municipal bond pricing and credit quality, with particular attention to electric utility infrastructure and its geographic vulnerabilities.

Methodology

We model bond spreads as functions of the Fire Weather Index (FWI) and historical fire occurrence. In this framework, the FWI reflects past atmospheric conditions derived from historical scenarios, while historical fire data capture the legacy of wildfire impacts. The interaction between these two variables allows us to assess whether areas with prior wildfire exposure exhibit increased sensitivity to elevated FWI levels.

Both equations also include a vector of control variables, as well as utility- and time-fixed effects, to isolate the impact of wildfire risk on the cost of capital and credit quality. Using the FWI enables us to build a predictive model of bond spreads under future climate scenarios.

$$\begin{aligned}\text{BondSpread}_{u,t} = \alpha + \beta_1 \text{FWI}_{u,t} + \beta_2 \text{HistoricFireOccur}_{u,t} + \beta_3 \left(\text{FWI}_{u,t} \times \text{HistoricFireOccur}_{u,t}\right) \\ + \gamma' \mathbf{X}_{u,t} + \delta_u + \theta_t + \varepsilon_{u,t}\end{aligned}$$

Where,

- u = utility or issuer index
- t = time index (e.g., year-month)
- $\text{BondSpread}_{u,t}$ = yield spread or cost of capital for the utility's bond
- $\text{FWI}_{u,t}$ = Fire Weather Index in the utility's service area and time t
- $\text{HistoricFireOccur}_{u,t}$ = indicator or severity measure of any historic fire in that area/time
- $\mathbf{X}_{u,t}$ = vector of controls (bond characteristics, demographics, etc.)
- δ_u = utility-issuer fixed effects
- θ_t = time fixed effects
- $\varepsilon_{u,t}$ = error term

In constructing our empirical framework, we draw inspiration from the factor selection methodology outlined by Jeon, Barrage, and Walsh [2], who examine the relationship between wildfire risk and school district bond spreads. Their model includes a wide range of control variables such as fiscal capacity, property wealth, enrollment size, and demographic characteristics, each representing dimensions that influence a district's revenue-generating

potential. By analogy, for electric utilities, an analogous set of factors would include service area wealth, customer base size, rate-setting autonomy, debt service coverage, and affordability thresholds.

These socioeconomic and operational factors are also emphasized in Moody's municipal utility revenue debt methodology, which highlights service area income levels, rate flexibility, and customer vulnerability as key indicators of a utility's revenue resilience [6]. Although our full model may be constrained by data availability, we aim to incorporate the most policy-relevant and observable dimensions of socioeconomic risk. Among these, we prioritize service area wealth, proxied by median household income, as our primary socioeconomic control variable. Wealthier service areas are likely to exhibit greater rate flexibility, resilience to rate shocks, and improved capacity to absorb wildfire-related capital costs, which can influence a utility's credit risk exposure.

In addition to socioeconomic variables, we control for bond-specific characteristics standard in municipal bond pricing analysis [2], including bond maturity (time until principal repayment), coupon rate (annual interest rate paid), callability (whether the bond can be redeemed early), amount outstanding (size of the issue), and insurance status (whether bond payments are guaranteed by a third party). These contractual features systematically affect yield spreads and liquidity and are essential for isolating the relationship between wildfire risk and borrowing costs.

To address unobserved heterogeneity across issuers, we incorporate utility-issuer fixed effects into our specification. These fixed effects control for all time-invariant characteristics of each utility, including baseline financial health, regulatory environment, managerial quality, and geographic location. Incorporating fixed effects ensures that the estimated impact of wildfire risk is not confounded by stable differences across utilities that could otherwise bias our results.

Results

To ensure the reliability of bond trade data used in the analysis, a series of filters are applied to remove outliers and non-standard observations. Trades with negative yields are excluded, as are those with dollar prices outside the \$50 to \$150 range, which helps eliminate potentially erroneous or unusually structured bonds. Bonds offering coupon rates greater than 20% are removed due to their atypical nature, and those with more than 30 years remaining until maturity are excluded to reduce long-horizon sensitivity. Additionally, bonds that have been traded fewer than 10 times are filtered out to maintain a baseline level of market liquidity and data robustness.

To isolate the short-run impact of climate-related disasters, particularly wildfires, the analysis focuses on trades that occurred within a 30-day window following the ignition of a fire. This approach is informed by the methodology developed by Jeon, Barrage, and Walsh in "*Pricing Climate Risks: Evidence from Wildfires and Municipal Bonds.*" [2]. By computing the number of days between the ignition date and each trade date, only transactions that fall within 0 to 30 days post-ignition are retained. This targeted time frame allows the study to capture pricing effects that are most likely attributable to the wildfire event, thereby limiting interference from unrelated market developments.

To handle missing values in the variable average adjusted gross income, we employ a targeted imputation strategy. For observations after 2022 or before 2014 with missing income values, we fill in the data using mapped values from income snapshots available in 2022 and 2014, respectively. This approach provides a forward-fill and backward-fill mechanism that maintains geographic consistency via NAME-based mapping, while preserving time-relevance for socio-economic variables.

Panel Regression Model Formulation

To assess the impact of wildfire risk and weather conditions on municipal bond spreads, we estimate the following fixed-effects panel regression model. This specification captures both the contemporaneous climate risk via the Fire Weather Index (FWI) and legacy effects of past wildfire exposure, while controlling for bond-specific and socioeconomic characteristics:

$$\begin{aligned}\text{BondSpread}_{u,t} = {} & \alpha + \beta_1 \text{Fire}_{u,t} + \beta_2 \text{FWI}_{u,t} + \beta_3 (\text{Fire}_{u,t} \times \text{FWI}_{u,t}) \\ & + \gamma_1 \text{ParTraded}_{u,t} + \gamma_2 \text{CallFlag}_{u,t} + \gamma_3 \text{InsuranceFlag}_{u,t} + \gamma_4 \text{Customers}_{u,t} + \gamma_5 \text{Income}_{u,t} \\ & + \gamma_6 \text{BurnedAcres}_{u,t} + \gamma_7 \text{DaysSinceIgnition}_{u,t} + \gamma_8 \text{YearsToMaturity}_{u,t} + \gamma_9 \text{Coupon}_{u,t} \\ & + \delta_u + \theta_t + \varepsilon_{u,t}\end{aligned}$$

Where,

- u = utility or issuer index
- t = time index (e.g., year-month)
- $\text{BondSpread}_{u,t}$ = yield spread or cost of capital for the utility's bond
- $\text{Fire}_{u,t}$ = indicator or measure of wildfire severity in the service area at time t (e.g., mean or max severity)
- $\text{FWI}_{u,t}$ = Fire Weather Index in the utility's service area at time t

- $\text{Fire}_{u,t} \times \text{FWI}_{u,t}$ = interaction term to capture conditional climate risk sensitivity
- $\text{ParTraded}_{u,t}$ = number of trades for the bond (proxy for liquidity)
- $\text{CallFlag}_{u,t}$ = indicator for whether the bond is callable
- $\text{InsuranceFlag}_{u,t}$ = indicator for whether the bond is insured
- $\text{Customers}_{u,t}$ = number of customers served by the utility
- $\text{Income}_{u,t}$ = average adjusted gross income in the service area
- $\text{BurnedAcres}_{u,t}$ = number of acres burned in the associated wildfire
- $\text{DaysSinceIgnition}_{u,t}$ = number of days since the wildfire ignition
- $\text{YearsToMaturity}_{u,t}$ = remaining years to maturity of the bond
- $\text{Coupon}_{u,t}$ = bond's coupon rate
- δ_u = utility or issuer fixed effects (controls for time-invariant utility characteristics)
- θ_t = time fixed effects (controls for macroeconomic conditions or market-wide shocks)
- $\varepsilon_{u,t}$ = idiosyncratic error term

Panel Regression Results and Interpretation

We estimate four panel regression models to assess the relationship between municipal bond spreads and wildfire-related risk factors, using different combinations of fire severity (`fire_sev_mean`, `fire_sev_max`) and Fire Weather Index (FWI) metrics (`fwi_mean`, `fwi_max`). The objective is to understand how physical climate risk—both as immediate threat and historical exposure—is priced into municipal credit markets.

Among the four models, those incorporating `fire_sev_mean` as the measure of fire severity consistently yield statistically significant results for the main variables of interest. In both the `fire_sev_mean + fwi_mean` and `fire_sev_mean + fwi_max` specifications, the coefficients on `fire`, `fwi`, and their interaction term (`fire` $\times$ `fwi`) are all significant at the 5% level. The direction of these coefficients provides valuable insight. The positive coefficients on both `fire` and `fwi` suggest that the bond market reacts to both actual wildfire occurrence and heightened fire weather risk by increasing the yield spreads demanded by investors. This indicates that physical wildfire risk—both realized and potential—is priced as a source of credit risk.

Interestingly, the coefficient on the interaction term (`fire` $\times$ `fwi`) is negative and significant. This suggests a dampening effect: when both fire severity and FWI are simultaneously high,

the marginal impact on bond spreads is smaller than when either variable is high on its own. One possible interpretation is that the market views a wildfire during extreme weather conditions (high FWI) as an uncontrollable event, and thus may not penalize the issuer as severely. Conversely, when a severe fire occurs during low-FWI conditions, the event may be interpreted as a failure of preparedness or operational negligence by the utility, leading to a stronger negative reaction in bond prices. This aligns with the hypothesis that investors differentiate between exogenous risk and risk that may stem from utility-level governance or risk management lapses.

Models using `fire_sev_max` as the fire severity measure do not yield statistically significant coefficients for these core explanatory variables. This may reflect the limitations of the `max` metric, which could be influenced by extreme outliers such as single high-severity pixels in the fire footprint. In contrast, `fire_sev_mean` better captures the average exposure and appears to be a more representative and reliable metric for modeling the impact of wildfires on bond pricing.

Several control variables also provide meaningful insights. The variable `DaysSinceIgnition` typically exhibits a negative and often significant coefficient, suggesting that the effect of a wildfire on bond spreads diminishes as more time passes since the ignition date. This implies a temporal concentration of perceived risk: investors react strongly in the immediate aftermath of an event, but the perceived premium gradually fades. `YearsToMaturity` is another consistently significant variable with a negative coefficient, indicating that longer-dated bonds may trade at lower spreads—possibly due to expectations of long-term recovery or interest rate smoothing effects. In contrast, `Coupon` shows a positive and often significant relationship with bond spreads, which is consistent with standard bond pricing behavior: higher-coupon bonds tend to offer higher yields.

Although overall R^2 values are relatively low (ranging from 0.0021 to 0.0085), this is not unexpected in panel regressions with many omitted variables due to data limitations. Importantly, the models still identify statistically significant relationships between bond spreads and key climate risk variables. These findings support the broader conclusion that wildfire severity, fire weather conditions, and their interaction play an important role in how municipal bond markets evaluate credit risk in the face of climate hazards.

Discussion

Our methodology includes several critical decisions that merit further reflection. First, we identified municipal bonds associated with public utility districts based on the “Issuer Name” field, rather than filtering by “Purpose Class” equal to “Public Utility District.” This choice was made to achieve greater precision and coverage, as issuer naming conventions more

reliably capture relevant entities than the sometimes inconsistent purpose classifications in the dataset.

Second, the spatial matching between municipal bond issuers and utility service areas relied on the use of large language models (LLMs) to perform name-based geometry alignment. While this approach enhanced efficiency, it also introduced the potential for mismatches. Despite implementing manual verification and applying a fuzzy matching function to assess match quality, we observed that not all matches returned perfect similarity scores. This indicates that some matches, although plausible, may not represent true geographic correspondence. For future applications, developing a more rigorous and transparent algorithmic matching process would strengthen data reliability and reproducibility.

Third, while our research focuses primarily on electric utilities, we recognize that the dataset may include both Electric Revenue Bonds and Water Revenue Bonds. As detailed in utility bond classifications, these two types of revenue bonds serve distinct purposes and are subject to different operational risks. Electric Revenue Bonds finance power-related infrastructure such as transmission lines and generation facilities, with risk factors including demand variability, regulatory changes, and integration of renewable energy. In contrast, Water Revenue Bonds fund infrastructure such as treatment plants and pipelines, and face risks associated with water consumption patterns, environmental compliance, and aging systems. Failing to distinguish between these two could introduce noise into our analysis, particularly when assessing the relationship between wildfire risk and bond pricing, as water utilities may face different exposure pathways and financial dynamics compared to electric utilities.

Moreover, the sparsity of municipal bond trading data and the limited availability of spread observations over the past two decades present additional challenges. Many municipal bonds are infrequently traded, leading to uneven coverage across issuers, regions, and time periods. As a result, our panel dataset may be biased toward more liquid or recently issued bonds, potentially limiting the generalizability of our findings. Furthermore, the relatively short 20-year time frame of available data may not capture the full range of fire events or long-term shifts in climate risk perception. These limitations may reduce the precision of estimated relationships and highlight the importance of interpreting our results in the context of data constraints.

These limitations underscore the need for careful data curation and methodological refinement in future iterations of this research. More granular classification and improved matching techniques could further enhance the accuracy of empirical findings and the robustness of policy implications.

References

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Table 1: Key Bond Characteristics Considered in Control Variables

Category	**Variable Example**	**Description**
Bond Characteristics	Maturity (Years to)	Time until principal repayment
Bond Characteristics	Coupon Rate (%)	Annual interest rate paid
Bond Characteristics	Call Feature (Indicator)	Whether the bond can be redeemed early by the issuer
Bond Characteristics	Amount Outstanding (Log)	Size of the bond issue
Bond Characteristics	Insured (Indicator)	Whether bond payments are guaranteed by a third-party insurer